Quiz 4: Linear maps – Solutions

Tutorial 4, Tuesday, 27th October, 2026

Exercise 1. (6 Points) Decide which of the following functions are linear maps. If a function is linear, determine its matrix. If it is not linear, give an explicit counterexample.

$$f : \mathbb{R}^3 \longrightarrow \mathbb{R}^2 \qquad g : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$
$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \longmapsto \begin{pmatrix} x_1 - 2x_3 \\ 3x_2 + x_3 \end{pmatrix}, \qquad \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \longmapsto \begin{pmatrix} x_1 + 2x_2 \\ x_2^2 \end{pmatrix}.$$

Solution. f **is linear.** For all $x \in \mathbb{R}^3$, $f(x) = \begin{pmatrix} x_1 - 2x_3 \\ 3x_2 + x_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 3 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$, i.e. $f(x) = Ax$ with $A \in \mathbb{R}^{2\times 3}$; every map $x \mapsto Ax$ is linear.

The matrix of f (columns $f(e_1), f(e_2), f(e_3)$): $\boxed{[f] = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 3 & 1 \end{pmatrix}}$

$\boxed{g \text{ is not linear.}}$

Explicit counterexample, e.g. $u = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $\lambda = 2$: $g(\lambda u) = g\begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0+4 \\ 2^2 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$, but $\lambda g(u) = 2\begin{pmatrix} 0+2 \\ 1^2 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$, so $g(\lambda u) \neq \lambda g(u)$.

Exercise 2. (4 Points) Let $F : \mathbb{R}^2 \to \mathbb{R}^3$ be a linear map with

$$F\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \\ -2 \end{pmatrix}, \qquad F\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix}.$$

Calculate $F\begin{pmatrix} 3 \\ 1 \end{pmatrix}$.

Solution. Write $\begin{pmatrix} 3 \\ 1 \end{pmatrix} = \alpha\begin{pmatrix} 2 \\ 1 \end{pmatrix} + \beta\begin{pmatrix} 1 \\ 1 \end{pmatrix}$: $2\alpha + \beta = 3$, $\alpha + \beta = 1 \Rightarrow \alpha = 2$, $\beta = -1$, i.e. $\begin{pmatrix} 3 \\ 1 \end{pmatrix} = 2\begin{pmatrix} 2 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

By linearity, $F\begin{pmatrix} 3 \\ 1 \end{pmatrix} = 2\,F\begin{pmatrix} 2 \\ 1 \end{pmatrix} - F\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 10 \\ -4 \end{pmatrix} - \begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix} = \boxed{\begin{pmatrix} 5 \\ 6 \\ -2 \end{pmatrix}}$